

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	01 Dec 2012	First Release	ONG KEAN SHENG
01	2 Oct 2016	Template changes & Content update	ONG KEAN SHENG
02	20 Oct 2017	Content Update	MUHAMMED SHAMEEM

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

Content

Tools Requirement.....	3
Material.....	3
Failure Identification.....	4
X Ray Camera Removal &Replacement Procedure.....	6

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

Tools Requirement

- Phillips screw driver / Allen Key
- Camera tools (2350-0010)



Figure 1 Camera Tool

Material

X ray camera (2210-0023)

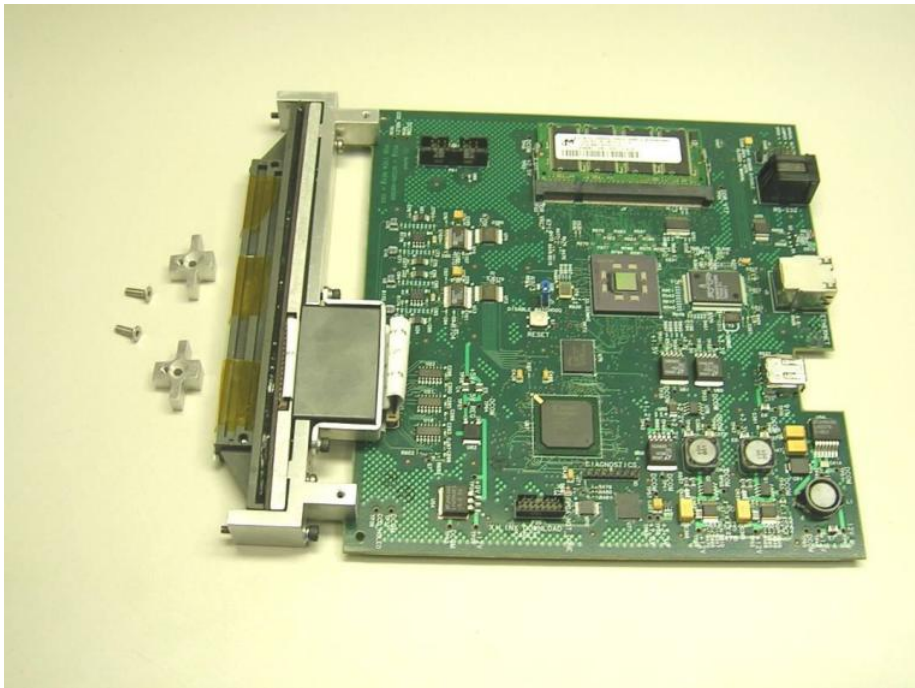
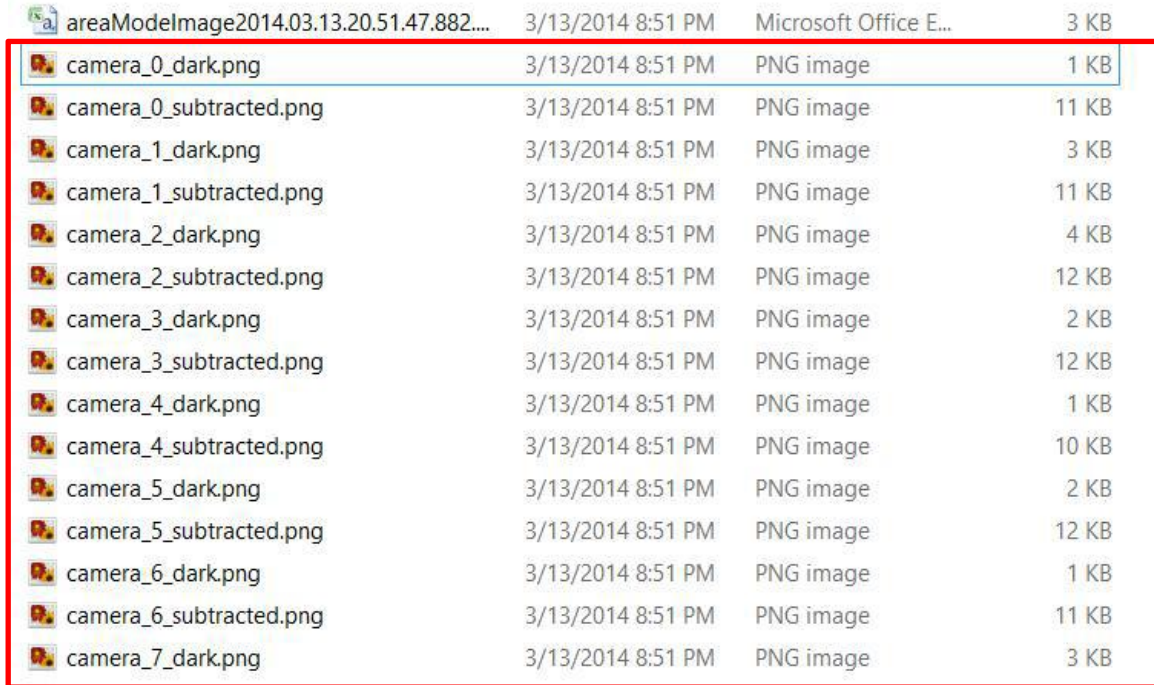


Figure 2 X ray Camera

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

Failure Identification

1. Go to "C:\Program Files\AXI System\v810\REV\ \log\confirm\areaModelImage.
2. Delete all entire .png file only except log file as below Figure 3.



areaModelImage2014.03.13.20.51.47.882....	3/13/2014 8:51 PM	Microsoft Office E...	3 KB
camera_0_dark.png	3/13/2014 8:51 PM	PNG image	1 KB
camera_0_subtracted.png	3/13/2014 8:51 PM	PNG image	11 KB
camera_1_dark.png	3/13/2014 8:51 PM	PNG image	3 KB
camera_1_subtracted.png	3/13/2014 8:51 PM	PNG image	11 KB
camera_2_dark.png	3/13/2014 8:51 PM	PNG image	4 KB
camera_2_subtracted.png	3/13/2014 8:51 PM	PNG image	12 KB
camera_3_dark.png	3/13/2014 8:51 PM	PNG image	2 KB
camera_3_subtracted.png	3/13/2014 8:51 PM	PNG image	12 KB
camera_4_dark.png	3/13/2014 8:51 PM	PNG image	1 KB
camera_4_subtracted.png	3/13/2014 8:51 PM	PNG image	10 KB
camera_5_dark.png	3/13/2014 8:51 PM	PNG image	2 KB
camera_5_subtracted.png	3/13/2014 8:51 PM	PNG image	12 KB
camera_6_dark.png	3/13/2014 8:51 PM	PNG image	1 KB
camera_6_subtracted.png	3/13/2014 8:51 PM	PNG image	11 KB
camera_7_dark.png	3/13/2014 8:51 PM	PNG image	3 KB

Figure 3 PNG File Delete

3. Open ConfigFolder by refer to below table :

Model	Path
Strd Series One & Series 2	C:\Program Files\AXI System\v810\5.0\config
XXL Series One & Series 2	C:\Program Files\AXI System\v810\5.0\XXLconfig
S2EX	C:\Program Files\AXI System\v810\5.0\s2exconfig

Table 1

4. Choose "Softwareconfig" and double click to open.
5. Change as " saveAllConfirmationImages=True"
6. Run one loop of CD & A.

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

7. Go to area mode image confirmation folder from "C:\Program Files\AXI System\v810\REV\
\log\confirm\areaModelImage" and check the camera segment picture from camera 0 ~ camera 14.

Notes: If bad segment picture is resulted during checking, please help to proceed camera replacement procedure.



Figure 4 Good Camera Segment



Figure 5 Bad Camera Segment

8. Swap defective camera with other good camera. (May refer to X Ray Camera Replacement Procedure which listed at below)
9. Repeat Step 1 to Step 9 to validate the defective camera status.

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

X Ray Camera Removal & Replacement Procedure

1. Turn off X-Ray source
2. Turn off Trigger Board Switch as below Figure 6

Remarks : The switch is located behind Power Distribution Unit (PDU)

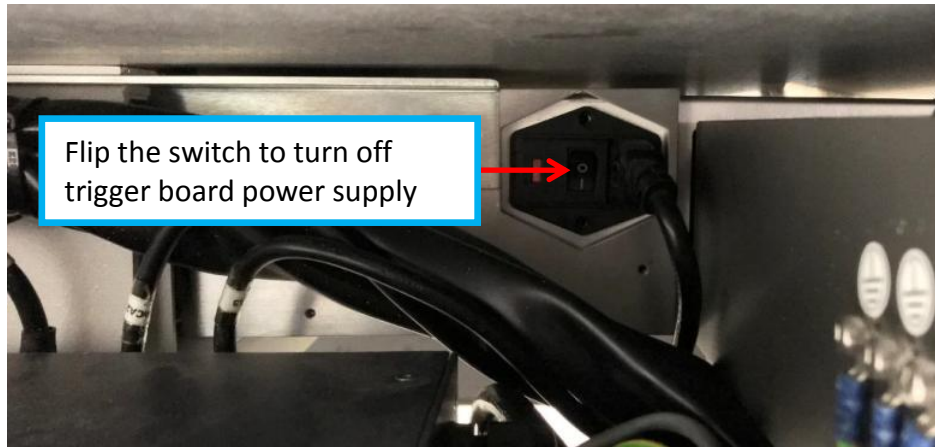


Figure 6 Trigger Board Switch

3. Unscrew 2 thumb screws that secure the camera enclosure as below Figure 7.



Figure 7 Camera Enclosure Thumb Screws

4. Slide the two doors of the camera array shield to open the camera enclosure as Figure 8.



Figure 8 Camera Array Opening

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

5. Disconnect the camera triggering cable (Firewire cable) and network cable from the defective X ray camera as below Figure 9.

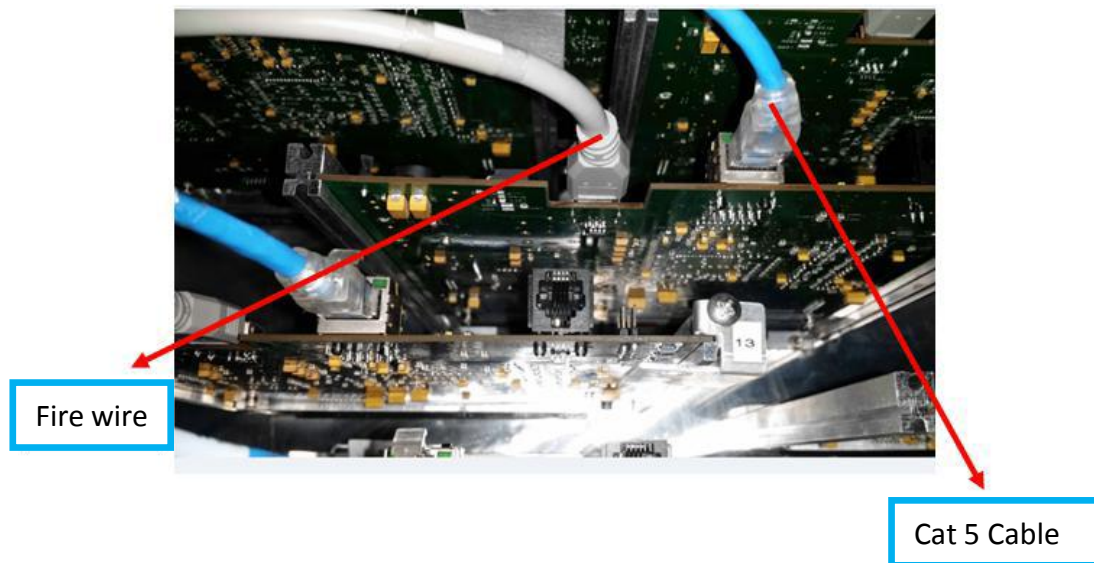


Figure 9 Firewire& Cat 5 Cable Removal

6. Use camera tools (2350-0010) to remove 2 x camera mounting screws as below Figure 10.



Left view mounting screw



Right view mounting screw

Figure 10 Left & Right Mounting Screws

VITrox	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

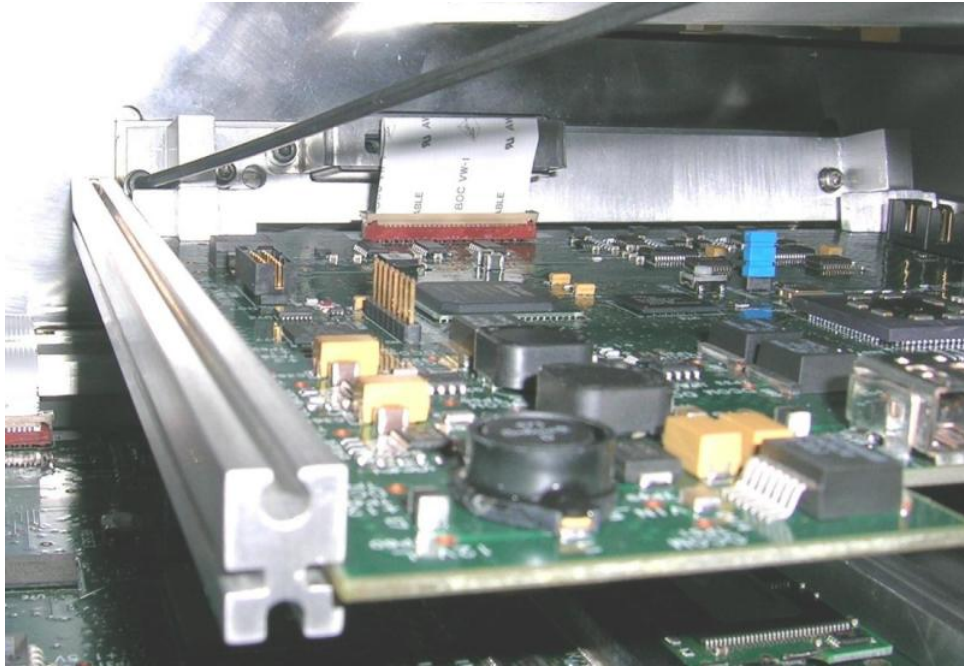


Figure 11 Camera removal (Reference Picture Only)

7. Remove the X ray camera from the slides
8. Install the new camera by gently slide into the camera slot accordingly
9. Beware friction between camera PCA `s component and camera flag during sliding
10. Tighten 2 x Camera mounting screws.
11. Open Command Prompt
12. Turn on trigger board switch and wait for 10 seconds
13. Type "Ping 192.168.128.128.XX" (May refer to below camera IP table for applying appropriate camera IP for particular camera) and check any return result as Figure 12.

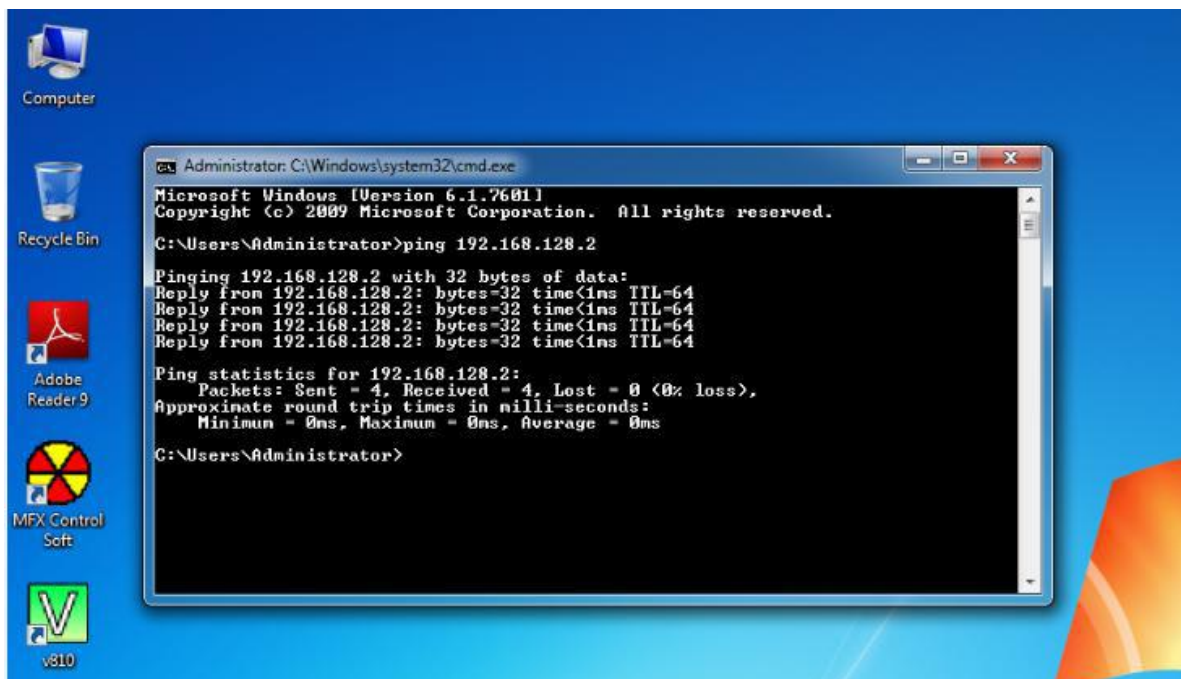


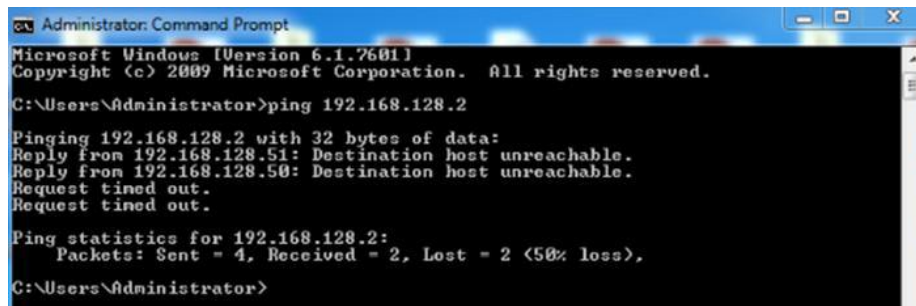
Figure 12 Command Prompt Ping Test

ViTrox	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

Camera ID	Camera IP
0	192.168.128.14
1	192.168.128.1
2	192.168.128.2
3	192.168.128.3
4	192.168.128.4
5	192.168.128.5
6	192.168.128.6
7	192.168.128.7
8	192.168.128.8
9	192.168.128.9
10	192.168.128.10
11	192.168.128.11
12	192.168.128.12
13	192.168.128.13

Table 2

a.If after ping test at command prompt and shown as below Figure 13, please help to proceed to Step b



```

Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.128.2

Pinging 192.168.128.2 with 32 bytes of data:
Reply from 192.168.128.51: Destination host unreachable.
Reply from 192.168.128.50: Destination host unreachable.
Request timed out.
Request timed out.

Ping statistics for 192.168.128.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\Administrator>

```

Figure 13 Ping Test Fail

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

- b. Adjust the camera flag (Camera number plate)'s until sensor able detect through to camera flag's tiny hole as Figure 13.

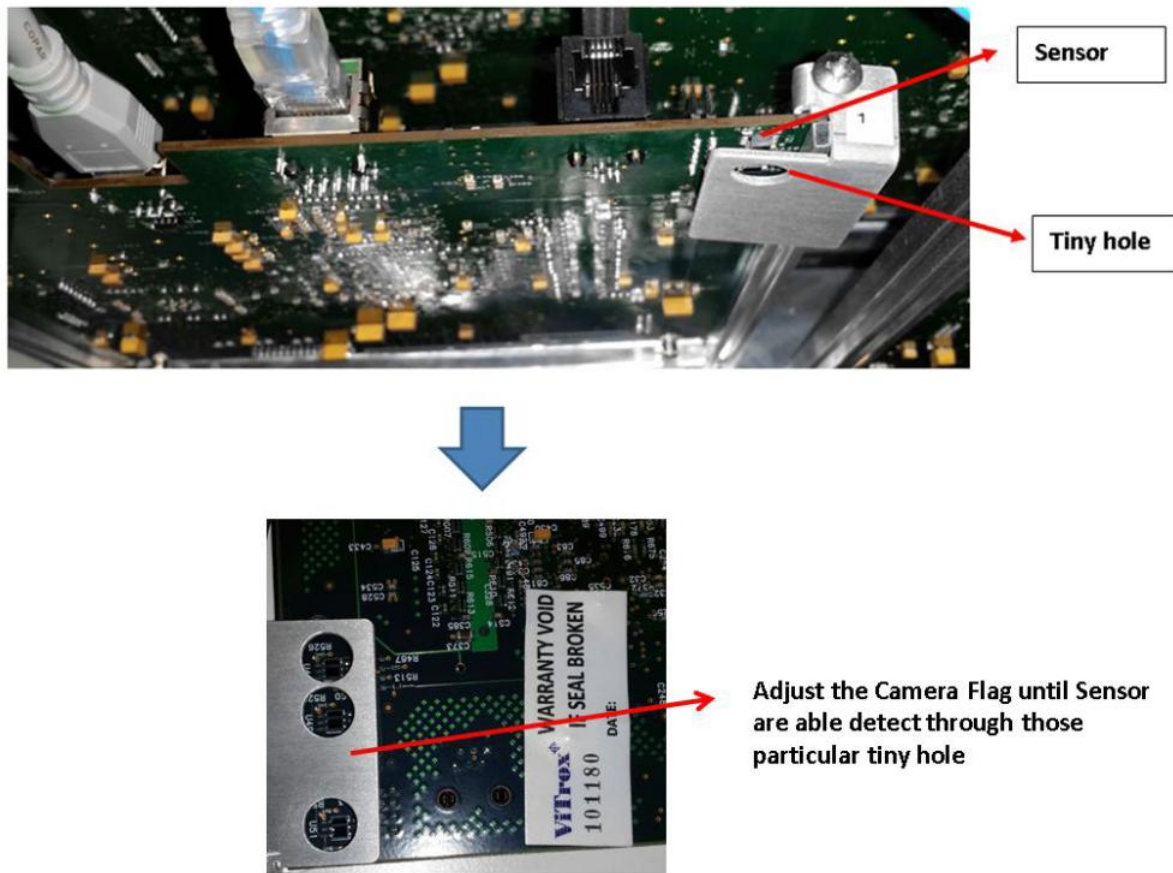


Figure 14 Camera Flag adjustment

Remarks:



Figure 15 Camera Flag Adjustment Distance (Reference Picture Only)

	Work Instruction	Doc. Revision	02
	AXI V810 X ray Camera Replacement Procedure	Effective Date	20 Oct 2017

- c. Reseat cat 5 cable and fire wire cable after adjust
- d. Perform Ping test as Step 13
- e. If Ping Test passes, proceed to Step 14.
- f. If Ping Test fail, repeat the Camera flag adjustment from Step a to d until Ping Test is pass

14. Close the camera enclosure.

15. Tighten 2x thumb screws

16. Click “ Upgrade firmware” at X ray Camera Tab (Under Service Mode) as below Figure 15

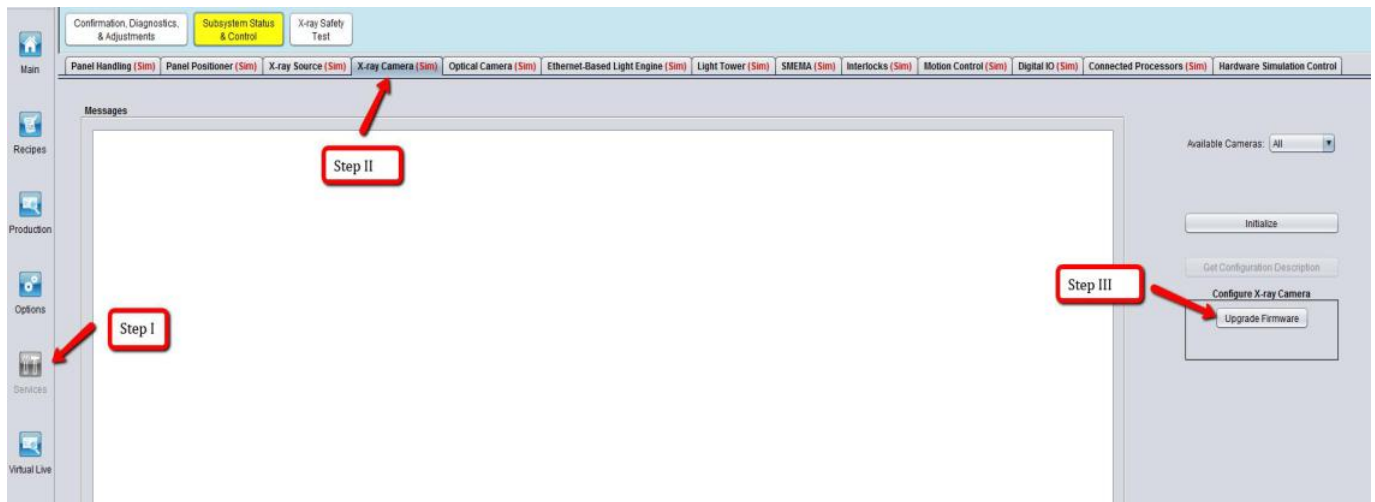


Figure 16 Camera Firmware Upgrade

17. Click “ Initialize” at X ray Camera Tab

18. Turn On X-Ray Source

19. Run 1 loop CD & A